

Marco Túlio Quintino

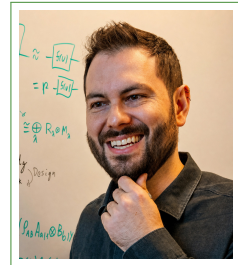
Curriculum Vitae (long version)

Associate Professor
(Maître de conférences HDR)
Sorbonne University, LIP6
Paris, France

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🌐 mtcq.github.io

Google Scholar, arXiv



Last updated: 6 October 2026

Academic profile

Sep 2022 – **Associate Professor (Maître de conférences HDR)**

Present Sorbonne University, Paris, France

Permanent member of the [Quantum Information](#) team at [LIP6](#)

Research interests: Quantum information and computation, quantum correlations (entanglement, Bell nonlocality, EPR steering, measurement incompatibility, and prepare-and-measure scenarios), quantum state and channel discrimination, higher-order quantum operations, causality in quantum theory, and semidefinite programming.

Selected highlights

- **Academic leadership.** Co-director of Sorbonne University's Master's programme in Quantum Information since September 2023.
- **Research output.** 68 research articles: 53 published or accepted journal articles and 15 preprints; more than 3700 citations ([Google Scholar](#), Oct 2026); more than 115 co-authors. [Full list](#), page 12.
- **Research supervision.** Current team: 1 postdoctoral researcher and 4 PhD students; completed co-supervision of 2 PhD theses and supervision or co-supervision of 5 master's theses.
- **Research funding.** Principal Investigator and coordinator of ANR JCJC HOQO-KS (2025–2030); supervisor of 4 PhD grants.
- **Teaching.** More than 150 hours of lectures and more than 410 hours of tutorials; created or substantially redesigned 2 master's courses in quantum information at Sorbonne University. [Teaching details](#), page 16.

Guide to this CV

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Education and employment

Education

- Jan 2025 **Habilitation in Computer Science (HDR)**, Sorbonne University, Paris, France
Quantum information processing via higher-order operations
 Defence date: 17 Jan 2025
 Jury: Pablo Arrighi, Anne Broadbent, Giulio Chiribella, Elham Kashefi, Mário Ziman
- Oct 2012 – **PhD in Physics**, University of Geneva, Geneva, Switzerland
 Sep 2016 *Quantum entanglement and measurement incompatibility as resources for nonlocality*
 Defence date: 9 Sep 2016
 Supervisor: Nicolas Brunner
 Funding: **SNF**, Switzerland
- Aug 2010 – **MSc in Physics**, Federal University of Minas Gerais (UFMG), Belo Horizonte, Brazil
 Sep 2012 *Black box correlations: locality, noncontextuality, and convex polytopes*
 Defence date: 12 Sep 2012
 Supervisor: Marcelo Terra Cunha, Co-supervisor: Daniel Cavalcanti
 Funding: **CAPES**, Brazil
- Feb 2007 – **BSc in Physics**, UFMG, Belo Horizonte, Brazil
 Aug 2010 *Não-localidade como recurso para comunicação*
 Supervisor: Marcelo Terra Cunha
 Funding: **CNPq**, Brazil

Previous positions

- Mar 2021 – **Postdoctoral researcher**, IQOQI Vienna, Vienna, Austria
 Aug 2022 Research group: Časlav Brukner
 Funding: **ESQ postdoc fellowship**
- Jun 2020 – **Postdoctoral researcher**, University of Vienna, Vienna, Austria
 Mar 2021 Research group: Časlav Brukner
 Funding: **Austrian Science Fund (FWF)** through the **SFB project BeyondC**
- Mar 2019 – **Postdoctoral researcher**, The University of Tokyo, Tokyo, Japan
 May 2020 Research group: Mio Murao
 Funding: **Q-LEAP project of the MEXT Japan**
- Nov 2016 – **Postdoctoral researcher**, The University of Tokyo, Tokyo, Japan
 Nov 2018 Research group: Mio Murao
 Funding: **JSPS postdoc fellowship**

Leadership and responsibilities

Programme leadership

- Sep 2023 – **Co-director** of the **Master's programme in Quantum Information**,
 Present Sorbonne University, Paris, France.
 Shared responsibility for student selection and admissions, curriculum design, and programme coordination. Member of the master's thesis examination committees for all students in the programme.

Recruitment panels

- 2024 – **National Institute of Informatics (NII) internship recruitment jury member**, Tokyo, Japan.
Evaluated and ranked LIP6 applicants for research internships at NII.
Calls: 2026 (**1st call**), 2025 (**1st** and **2nd call**), 2024 (**2nd call**).
- 2026 **QuanG2 PhD recruitment jury member**, Grenoble, France.
Evaluated and ranked candidates for the **Autumn 2026 PhD call** of the Quantum Grenoble doctoral programme in quantum sciences and technologies.
- 2025 **EDITE PhD recruitment jury member**, Paris, France.
Evaluated and ranked candidates for PhD funding on the **EDITE** Operations Research, Algorithms, Computing and Programming panel, chaired by Antoine Genitrini.

Research funding and fellowships

Research grants

- Sep 2025 – **ANR JCJC HOQO-KS (CE47-6396)**
Aug 2030 ANR (France): **EUR 358,660**
Principal Investigator and project coordinator
Higher-Order Quantum Operations with Known States
- Jan 2025 – **PHC Polonium (53077ZD; BPN/BFR/2024/1/00026)**
Dec 2026 MEAE and MESRI (France), and NAWA (Poland): **EUR 6,200** (French allocation)
Research mobility project, together with Michał Studziński
Efficient methods for storing and transforming quantum programs
- Jan 2025 – **MFBNA (2023.12518.PEX)**
Jan 2026 FCT (Portugal): **EUR 50,000** (total project budget)
Project team member; local coordinator: Ricardo Faleiro
The many face(t)s of Bell nonlocality and their applications
- Apr 2024 – **Tremplins nouveaux entrants & nouvelles entrantes**
Dec 2024 Sorbonne University (France): **EUR 5,000**
Principal Investigator
Higher-Order Quantum Operations with Known States (HOQO-KS)
- Nov 2016 – **Grant-in-Aid for JSPS Fellows (KAKENHI 16F16769)**
Mar 2019 JSPS (Japan): **JPY 2,100,000** in direct-cost funding
Co-Investigator and Foreign JSPS Fellow, with Mio Murao (PI)
Entanglement and nonlocality in quantum theory

Postdoctoral fellowships

- Mar 2021 – **ESQ Postdoc Fellowship Program**
Aug 2022 BMBWF (Austria) and European Union: **EUR 124,506**
Higher-Order Quantum Operations: Foundations and Applications
- Nov 2016 – **JSPS Postdoctoral Fellowship for Research in Japan (Standard)**
Nov 2018 JSPS (Japan): **JPY 8,888,000**

Student stipend

- 2010 – **CAPES Master's stipend**
 2012 CAPES (Brazil): **BRL 29,100**
Black box correlations: locality, noncontextuality, and convex polytopes

Funding for doctoral projects

- Dec 2025 – **Partenariats stratégiques et thématiques**
 Nov 2028 Sorbonne University: approximately **EUR 125,000** for a 36-month doctoral contract.
Polina Chernyshova, selected following an interview with the programme's selection panel.
 Double-degree PhD programme with The University of Tokyo
Higher-order quantum computing with partial information
- Nov 2025 – **DIM QuanTiP doctoral grant (QCDSSR)**
 Oct 2028 Île-de-France Region: approximately **EUR 125,000** for a 36-month doctoral contract.
Mariia Kvashchuk, selected following an interview with the programme's selection panel.
Quantum communication and discrimination in the single-shot regime
- Apr 2025 – **QuanTEdu France doctoral contract**
 Mar 2028 French National Quantum Plan: approximately **EUR 125,000** for a 36-month doctoral contract, plus **EUR 10,000** in research support.
Lucas E. A. Porto, selected following an interview with the programme's selection panel.
The relationship between quantum entanglement and Bell nonlocality
- Oct 2023 – **QuanTEdu France doctoral contract**
 Sep 2026 French National Quantum Plan: approximately **EUR 125,000** for a 36-month doctoral contract, plus **EUR 10,000** in research support.
Vanessa Brzić, selected following an interview with the programme's selection panel.
Higher-Order Quantum Operations: Foundations and Applications

Research supervision

- **Current team:** 1 postdoctoral researcher and 4 PhD students.
- **Completed:** co-supervision of 2 PhD theses; supervision of 3 master's theses; co-supervision of 2 master's theses; supervision of 5 short-term internships, including 1 undergraduate internship.

Postdoctoral researchers

- Aug 2026 – **Carlos Vieira**, Sorbonne University, Paris, France
 Present Funded by: FAPESP BEPE postdoctoral fellowship, grant 2025/27304-7.

PhD supervision

- Dec 2025 – **Polina Chernyshova**, Sorbonne University, Paris, France
 Present *Higher-order quantum computing with partial information*
 Co-supervisor: Mio Murao, The University of Tokyo (Tokyo, Japan)
- Nov 2025 – **Mariia Kvashchuk**, Sorbonne University, Paris, France
 Present *Quantum communication and discrimination in the single-shot regime*
 Co-supervisor: Akihito Soeda, NII (Tokyo, Japan)
- Apr 2025 – **Lucas E. A. Porto**, Sorbonne University, Paris, France
 Present *The relationship between quantum entanglement and Bell nonlocality*
- Oct 2023 – **Vanessa Brzić**, Sorbonne University, Paris, France
 Present *Higher-Order Quantum Operations: Foundations and Applications*

PhD co-supervision

- Jun 2017 – **Marcello Nery**, UFMG, Belo Horizonte, Brazil
 Jan 2022 *Non-classical common-cause and direct-cause*
 Main supervisor: Reinaldo Oliveira Vianna
 PhD completion date: 21 Jan 2022
- Jan 2018 – **Qingxiuxiong Dong**, The University of Tokyo, Tokyo, Japan
 Mar 2021 *Quantum algorithms for higher-order quantum transformations of universal unitary operations*
 Main supervisor: Mio Murao
 PhD completion date: 19 Mar 2021

PhD internship hosting

- Mar 2024 – **Ties-A. Ohst**, Sorbonne University, Paris, France
 Apr 2024 *Quantum memory in causally ordered processes of universal unitary operations*
 Main supervisor: Otfried Gühne
 PhD completion date: 21 Aug 2025

Master's thesis supervision

- Sep 2025 – **Kristin Sundal Lien**, Sorbonne University, Paris, France
 Aug 2026 *Entanglement and channel discrimination: sometimes less is more*
- Mar 2025 – **Charbel Eid**, Sorbonne University, Paris, France
 Sep 2025 *Optimal discrimination between quantum measurements and quantum instruments*
- Feb 2025 – **Loïc Divry-Kennedy**, Sorbonne University, Paris, France
 Aug 2025 *Discriminating qubit states and channels from any finite group*

Master's thesis co-supervision

- Jul 2018 – **Wataru Yokojima**, The University of Tokyo, Tokyo, Japan
 Feb 2020 *Consequences of preserving reversibility in quantum superchannels*
 Main supervisor: Mio Murao
- Jun 2017 – **Qingxiuxiong Dong**, The University of Tokyo, Tokyo, Japan
 Jan 2018 *Quantum implementability of maps and supermaps*
 Main supervisor: Mio Murao

Short-term internship supervision

- Feb 2024 – **Yuki Koizumi**, Sorbonne University, Paris, France
 Mar 2024 Undergraduate internship supervision
Equivalence between unitary estimation and deterministic port-based teleportation
- Jan 2024 – **Shijun Zhang**, Sorbonne University, Paris, France
 Apr 2024 Master's internship supervision
Perfect discrimination of unitary operations when k calls are available (continuation)
- Apr 2023 – **Shijun Zhang**, Sorbonne University, Paris, France
 Jun 2023 Master's internship supervision
Perfect discrimination of unitary operations when k calls are available
- Apr 2023 – **Charbel Eid**, Sorbonne University, Paris, France
 Jun 2023 Master's internship supervision
Optimal discrimination between quantum measurements and instruments
- Apr 2023 – **Ethan Obadia**, Sorbonne University, Paris, France
 Jun 2023 Master's internship supervision
Quantum channel discrimination with partially entangled states

Academic service

Workshop organisation

- Dec 2026 **Japanese-French Quantum Information 2026 Workshop (JFQI 2026)**, Tokyo, Japan
Lead organiser, in collaboration with Jessica Bavaresco
Forthcoming.
- Dec 2023 **Japanese-French Quantum Information 2023 Workshop (JFQI 2023)**, Tokyo, Japan
Lead organiser, in collaboration with Akihito Soeda

Conference programme committees

- Jun 2026 **Causalworlds 2026**, Grenoble, France
- Oct 2025 **YQIS 2025**, Barcelona, Spain
- Aug 2025 **AQIS 2025**, The University of Hong Kong (HKU), Hong Kong
- Jun 2025 **CEQIP 2025**, Smolenice Castle, Slovakia
- Sep 2024 **Causalworlds 2024**, Waterloo, Canada
- Aug 2024 **AQIS 2024**, Sapporo, Japan
- Sep 2019 **YQIS 2019**, Gdańsk and Sopot, Poland
- Sep 2018 **YQIS 2018**, Vienna, Austria

PhD thesis reviewer

- 25 Jun 2026 **Oskar Szymon Słowik**, Center for Theoretical Physics PAS, Warsaw, Poland
Spectral gaps, t-designs and ϵ -nets in quantum computing
Supervisors: Adam Sawicki and Oliver Reardon-Smith
- 19 Mar 2026 **José Nogueira**, Unicamp, Campinas, Brazil
Postquantum correlations and their consequences: From exclusivity graphs to Prepare-and-Measure scenarios
Supervisors: Rafael Rabelo and Marcelo Terra Cunha
- 29 Oct 2025 **Peter Sidajaya**, National University of Singapore, Singapore
Studies in quantum foundations
Supervisor: Valerio Scarani
- 6 Oct 2025 **Pierre Pocreau**, Université Grenoble Alpes, Grenoble, France
The use of entanglement in non-cooperative games, and the computational power of indefinite causal order
Supervisors: Mehdi Mhalla and Alastair Abbott
- 16 May 2025 **Carlos de Gois**, University of Siegen, Siegen, Germany
Certifying quantum states and their properties through correlations, communication and tomography
Supervisor: Otfried Gühne
- 17 Dec 2024 **Lucas Pollyceno**, Unicamp, Campinas, Brazil
Multipartite Bell nonlocality and communication
Supervisor: Rafael Rabelo

- 23 Oct 2023 **Carles Roch i Carceller**, Department of Physics, Technical University of Denmark, Copenhagen, Denmark
Quantum state discrimination with applications in contextuality and randomness certification
Supervisor: Jonatan Bohr Brask
- 12 Jun 2023 **Jef Pauwels**, Université libre de Bruxelles, Brussels, Belgium
Entanglement and quantum communication between partially characterized devices
Supervisor: Stefano Pironio
- 10 Mar 2023 **Carlos Vieira**, Unicamp, Campinas, Brazil
Exploring the role of entanglement in sets of behaviors from prepare-and-measure scenarios
Supervisor: Marcelo Terra Cunha

PhD examination committee member

- 29 Sep 2026 **Ezra Acalapati**, Sorbonne University, Paris, France
Geometric and information-theoretic structures in quantum systems: complexity, control, and complementarity
Supervisor: Giuseppe Policastro
- 3 Feb 2022 **Roberto D. Baldijão**, Unicamp, Campinas, Brazil
Quantum Darwinism and Contextuality
Supervisor: Marcelo Terra Cunha

Master's thesis reviewer

- 31 Jan 2024 **Arthur C. R. Dutra**, Unicamp, Campinas, Brazil
Can KS-Contextuality Hide in a Crowd? Investigating state-independent contextuality in systems with multiple observers
Supervisor: Marcelo Terra Cunha
- 9 Apr 2021 **Lucas Pollyceno**, Unicamp, Campinas, Brazil
Novos critérios para o princípio da causalidade de informação
Supervisor: Marcelo Terra Cunha

Doctoral progress and monitoring committees (Comité individuel de suivi)

- Sep 2026 **Thales Gottardello Marrafon**, SOKENDAI, NII, Tokyo, Japan
Quantum operations over quantum network: resource analysis and optimization
Supervisor: Akihito Soeda
- Jul 2026 **Amanda M. Fonseca**, Université Grenoble Alpes, Grenoble, France
Sep 2025 *Characterising causal structures: classical, quantum, and relativistic causality*
Supervisors: Cyril Branciard and V. Vilasini
- Jun 2026 **Youness Ayaita**, Sorbonne University, Paris, France
Sep 2025 *Le raisonnement ampliatif : Expliquer le succès épistémique de la science*
Oct 2024 Supervisor: Anouk Barberousse
- Jun 2026 **Mu-En Liu**, Institut de Physique Théorique (IPhT), Gif-sur-Yvette, France
Tsirelson Inequalities in the Minimal Bell Scenario
Supervisor: Jean-Daniel Bancal
- May 2025 **Octave Mestoudjian**, Université Paris-Saclay, Palaiseau, France
Jul 2024 *Sous-systèmes généralisés en informatique quantique*
Supervisors: Pablo Arrighi and Augustin Vanrietvelde; also Mathew Wilson in 2025

Jul 2024 **Pierre Pocreau**, Université Grenoble Alpes, Grenoble, France
 Jun 2023 *Implications of causal indefiniteness for quantum communication*
 Supervisors: Mehdi Mhalla and Alastair Abbott

Peer review

- **Journals:** Quantum, PRL, PRX, PRX Quantum, PRA, Phys. Rev. Research, J. Math. Phys., Commun. Math. Phys., NJP, J. Phys. A, npj Quantum Information, and Optica.
- **Conferences:** QIP, STOC, TQC, AQIS, and YQIS.

Conferences, schools, and seminars

- **Invited talks:** 14 invitations, including 2 to events cancelled due to COVID-19.
- **Contributed talks:** 22 accepted talks at 21 conferences, including 3 accepted at QIP.
- **Summer schools:** invited lecturer at 1 summer school.

Invited talks

Sep 2026 **Workshop on the Structure of Quantum Theory**, Münster, Germany
The most discriminable quantum states in the multicopy regime

Aug 2026 **YQIS 2026**, Vienna, Austria
Optimal pure state cloning and transposition are complementary channels

Apr 2026 **851st WE-Heraeus Seminar: 90 Years of Einstein–Podolsky–Rosen Paradox**, Physikzentrum Bad Honnef, Germany
Understanding the relationship between measurement incompatibility, EPR steering, and Bell nonlocality

Feb 2026 **Quantum Correlations and Measurements**, Les Diablerets, Switzerland
Optimal pure state cloning and optimal pure state transposition are complementary channels

Jul 2025 **CEQIP 2025**, Smolenice Castle, Slovakia
Quantum information processing via higher-order operations

Jun 2025 **FoQaCiA Workshop 2025**, Braga, Portugal
Can the quantum switch be deterministically simulated?

Apr 2025 **BQIT 2025**, Bristol, UK
Can the quantum switch be deterministically simulated?

Apr 2022 **Tsirelson Memorial Workshop**, Vienna, Austria
Measurement incompatibility and Bell nonlocality: from 1985 to 2022

Mar 2020 **Second Workshop on Quantum Protocols**, Waseda University, Tokyo, Japan
 Workshop cancelled due to the COVID-19 pandemic
Reversing Unknown Quantum Transformations: Universal Quantum Circuit for Inverting General Unitary Operations

Mar 2020 **The Order of Things (TOOT)**, Obergurgl, Austria
 Conference cancelled due to the COVID-19 pandemic

Jan 2020 **Quantum Information Structure of Spacetime (QISS 2020)**, Hong Kong
Reversing unknown quantum transformations: A universal quantum circuit for inverting general unitary operations

Nov 2018 **Quantum Maiwar**, Brisbane, Australia
Semi-device-independent certification of indefinite causal order

- Sep 2018 **Post-AQIS 2018 Workshop**, Nagoya, Japan
Reversing unknown quantum transformations: A universal protocol for inverting general unitary operations
- Aug 2017 **FQXi Workshop: Quantum Incompatibility**, Laach Lake, Germany
Genuine n -wise measurement incompatibility and device-independent certificates of incompatibility

Summer school

- Aug 2026 **QALYPSO 2026: Quantum Optimisation Techniques & Physics of Computation**, Valletta, Malta
Invited speaker and advanced-track lecturer: *SDP as a tool for quantum information*—2 lectures: *What is an SDP?* and *Dual certificates, strong duality, and numerical SDPs*.

Contributed talks

- Aug 2026 **Quantum Physics and Logic 2026 (QPL 2026)**, Amsterdam, The Netherlands
Composable simultaneous purification: when all communication scenarios reduce to spatial correlations
- Jul 2025 **Quantum Physics and Logic 2025 (QPL 2025)**, Varna, Bulgaria
One-to-one correspondence between deterministic port-based teleportation and unitary estimation
- Jul 2025 **Quantum Physics and Logic 2025 (QPL 2025)**, Varna, Bulgaria
All incompatible measurements on qubits lead to multiparticle Bell nonlocality
- Feb 2025 **Quantum Information Processing 2025 (QIP 2025)**, Raleigh, United States
One-to-one correspondence between deterministic port-based teleportation and unitary estimation
(presented by Satoshi Yoshida)
- Aug 2023 **Asian Quantum Information Science 2023 (AQIS 2023)**, Seoul, South Korea
Simulating qubit correlations with classical communication
- Feb 2023 **Quantum Information Processing 2023 (QIP 2023)**, Ghent, Belgium
Simulating qubit correlations with classical communication
(presented by Martin J. Renner)
- Sep 2021 **SFB BeyondC Autumn Workshop 2021**, Innsbruck, Austria
Universal protocols for transforming unitary quantum operations
- Jun 2021 **Quantum Physics and Logic 2021 (QPL 2021)**, Gdańsk (online), Poland
Success-or-draw: A strategy allowing repeat-until-success in quantum computation
- Jan 2020 **Quantum Information Processing 2020 (QIP 2020)**, Shenzhen, China
Adaptive circuits exponentially outperform parallel ones for universal unitary inversion
- Aug 2019 **Asian Quantum Information Science 2019 (AQIS 2019)**, Seoul, South Korea
Semi-device-independent certification of indefinite causal order
- Sep 2018 **Asian Quantum Information Science 2018 (AQIS 2018)**, Nagoya, Japan
Reversing unknown quantum transformations
- Jul 2018 **Modern Topics in Quantum Information**, Natal, Brazil
Reversing unknown quantum transformations
- May 2017 **36th Quantum Information Technology Symposium (QIT36)**, Kyoto, Japan
Super-activation of quantum steering
- Mar 2016 **Quantum Networks (FQXi)**, Barcelona, Spain
Entangled states with a local hidden-variable model for sequential measurements

- Dec 2015 **Quantum Correlations, Contextuality and All That... Again**, Natal, Brazil
Algorithmic construction of local hidden variable models for entangled quantum states
- Aug 2015 **V Quantum Information School and Workshop**, Paraty, Brazil
Joint measurability, EPR steering, and Bell nonlocality
- Aug 2014 **Asian Quantum Information Science 2014 (AQIS 2014)**, Kyoto, Japan
Joint measurability, EPR steering, and Bell nonlocality
- Dec 2013 **Quantum Correlations, Contextuality and All That**, Natal, Brazil
Measurement Incompatibility in Quantum Mechanics
- Aug 2013 **IV Quantum Information School and Workshop**, Paraty, Brazil
Genuine Hidden Quantum Nonlocality
- Jun 2012 **Workshop on Quantum Correlations**, Natal, Brazil
All noncontextuality inequalities for the n -cycle scenario
- Dec 2011 **III Encontro temático do INCT-IQ**, Natal, Brazil
From the detection loophole to the transmission loophole
- Aug 2011 **III Quantum Information School and Workshop**, Paraty, Brazil
Maximal CHSH violations with low efficiency photodetection and homodyne measurements

Invited seminars

- 30 Apr 2025 **University of Siegen**, Siegen, Germany
Host: Otfried Gühne
- 12 Dec 2024 **The University of Hong Kong (HKU)**, Hong Kong
Host: Giulio Chiribella
- 11 Nov 2024 **University of Geneva**, Geneva, Switzerland
Host: Nicolas Brunner
- 8 Jul 2024 **Université Grenoble Alpes**, Grenoble, France
Host: Alastair Abbott
- 25 Jun 2024 **University of Pisa**, Pisa, Italy
Host: Costantino Budroni
- 24 Oct 2023 **Technical University of Denmark, QPIT**, Copenhagen, Denmark
Host: Jonatan Bohr Brask
- 16 Mar 2023 **Les Ateliers du LKB, Laboratoire Kastler Brossel**, Paris, France
Host: Nancy Paul
- 10 Dec 2021 **University of Gdańsk**, Gdańsk, Poland
Host: Michał Studziński
- 18 Mar 2021 **Quantin Research Group**, Warsaw (online), Poland
Host: Michał Oszmaniec
- 5 Feb 2021 **Technical University of Denmark**, Copenhagen (online), Denmark
Host: Jonatan Bohr Brask
- 10 Jul 2019 **Universitat Autònoma de Barcelona**, Barcelona, Spain
Host: Andreas Winter
- 14 Feb 2019 **ICFO**, Barcelona, Spain
Host: Antonio Acín
- 12 Feb 2019 **University of Geneva**, Geneva, Switzerland
Host: Nicolas Brunner

24 Jan 2019	IQOQI Vienna , Vienna, Austria Host: Marcus Huber
16 Dec 2016	The University of Tokyo , Tokyo, Japan Host: Mio Murao
18 Nov 2016	UFMG , Belo Horizonte, Brazil Host: Marcelo Terra Cunha
18 Nov 2015	University of Siegen , Siegen, Germany Host: Otfried Gühne
18 Aug 2015	UFMG , Belo Horizonte, Brazil Host: Marcelo Terra Cunha
31 Jul 2015	UFMG , Belo Horizonte, Brazil Host: Marcelo Terra Cunha
15 Jun 2015	Waseda University , Tokyo, Japan Host: Kazuya Yuasa
26 May 2015	Perimeter Institute (PI) , Waterloo, Canada Host: Matthew Pusey
9 Oct 2014	ICFO , Barcelona, Spain Host: Antonio Acín
5 Sep 2014	KCIK , Gdańsk, Poland Host: Michał Horodecki
25 Aug 2014	The University of Tokyo , Tokyo, Japan Host: Mio Murao
3 Dec 2013	UFMG , Belo Horizonte, Brazil Host: Marcelo Terra Cunha
19 Jul 2012	KCIK , Gdańsk, Poland Host: Michał Horodecki
11 Jul 2012	ICFO , Barcelona, Spain Host: Antonio Acín

Outreach and dissemination

During my JSPS postdoctoral fellowship in Japan, I presented my research to high-school students through the [JSPS Science Dialogue Program](#).

21 Sep 2018	JSPS Science Dialogue , Suwa Seiryō High School (Nagano Prefecture), Japan <i>What do you mean by random? Randomness, Quantum Mechanics, and Bell Non-locality</i>
5 Dec 2017	JSPS Science Dialogue , Takezono High School (Ibaraki Prefecture), Japan <i>What do you mean by random? Randomness, Quantum Mechanics, and Bell Non-locality</i>
7 Jan 2013	Science Slam Berlin , SO36, Berlin, Germany <i>Does God play dice? Genuine randomness in nature</i>

Languages and computational skills

- **Languages:** Portuguese (native), English (fluent), French (fluent), Spanish (advanced), and Japanese (intermediate).
- **Programming:** MATLAB and Mathematica (advanced); familiarity with Julia, Python, and Bash.
- **Research code:** Research-related computational code shared on [GitHub](#).

List of publications

All my scientific publications are available on [arXiv](#) and my [personal website](#). Publications since I started my position in France in September 2022 are also available on [HAL](#).

Peer-reviewed publications

- [1] L. E. A. Porto, S. Designolle, S. Pokutta, and **M. T. Quintino**. “Measurement incompatibility and quantum steering via linear programming”. *Quantum* **10**, 2141 (2026).
- [2] P. Taranto, S. Milz, M. Murao, **M. T. Quintino**, and K. Modi. “A Guide to Higher-order quantum operations”. *PRX Quantum* (2026). Accepted for publication. [arXiv:2503.09693 \[quant-ph\]](#).
- [3] F. Grosshans, M. Horodecki, M. Murao, T. Młynik, **M. T. Quintino**, M. Studziński, and S. Yoshida. “Multicopy quantum state teleportation with application to storage and retrieval of quantum programs”. *Quantum* **10**, 2105 (2026).
- [4] S. Yoshida, Y. Koizumi, M. Studziński, **M. T. Quintino**, and M. Murao. “One-to-one correspondence between deterministic port-based teleportation and unitary estimation”. *IEEE Trans. Inf. Theory* **72**, 2358–2377 (2026).
- [5] T.-A. Ohst, S. Zhang, H. C. Nguyen, M. Plávala, and **M. T. Quintino**. “Characterising memory in quantum channel discrimination via constrained separability problems”. *Quantum* **10**, 1988 (2026).
- [6] J. Bavaresco, H. Kristjánsson, M. Murao, T. Otake, **M. T. Quintino**, P. Taranto, and S. Yoshida. “Simulating the quantum switch with quantum circuits is computationally hard”. *Nat. Commun.* **16**, 10216 (2025).
- [7] M. Plávala, O. Gühne, and **M. T. Quintino**. “All incompatible measurements on qubits lead to multiparticle Bell nonlocality”. *Phys. Rev. Lett.* **134**, 200201 (2025).
- [8] S. Egelhaaf, J. Pauwels, **M. T. Quintino**, and R. Uola. “Certifying measurement incompatibility in prepare-and-measure and Bell scenarios”. *J. Phys. A: Math. Theor.* **58**, 095304 (2025).
- [9] S. Milz and **M. T. Quintino**. “Characterising transformations between quantum objects, ‘completeness’ of quantum properties, and transformations without a fixed causal order”. *Quantum* **8**, 1415 (2024).
- [10] P. Taranto, **M. T. Quintino**, M. Murao, and S. Milz. “Characterising the hierarchy of multi-time quantum processes with classical memory”. *Quantum* **8**, 1328 (2024).
- [11] T. Strömberg, P. Schiinsky, **M. T. Quintino**, M. Antesberger, L. A. Rozema, I. Agresti, Č. Brukner, and P. Walther. “Experimental superposition of a quantum evolution with its time reverse”. *Phys. Rev. Research* **6**, 023071 (2024).
- [12] L. Villegas-Aguilar, E. Polino, F. Ghafari, **M. T. Quintino**, K. T. Laverick, I. R. Berkman, S. Rogge, L. K. Shalm, N. Tischler, E. G. Cavalcanti, S. Slussarenko, and G. J. Pryde. “Nonlocality activation in a photonic quantum network”. *Nat. Commun.* **15**, 3112 (2024).

- [13] M. Antesberger, **M. T. Quintino**, P. Walther, and L. A. Rozema. “Higher-order process matrix tomography of a passively-stable quantum switch”. *PRX Quantum* **5**, 010325 (2024).
- [14] Q. Dong, **M. T. Quintino**, A. Soeda, and M. Murao. “The quantum switch is uniquely defined by its action on unitary operations”. *Quantum* **7**, 1169 (2023).
- [15] M. J. Renner and **M. T. Quintino**. “The minimal communication cost for simulating entangled qubits”. *Quantum* **7**, 1149 (2023).
- [16] T. Strömberg, P. Schiansky, R. W. Peterson, **M. T. Quintino**, and P. Walther. “Demonstration of a quantum SWITCH in a Sagnac configuration”. *Phys. Rev. Lett.* **131**, 060803 (2023).
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Theses and monographs

- [69] **M. T. Quintino**. *Quantum information processing via higher-order operations*. Habilitation thesis (HDR). 2025. URL: https://mtcq.github.io/assets/thesis/HDR_MTQ_final_version.pdf.
- [70] **M. T. Quintino**. *Quantum entanglement and measurement incompatibility as resources for nonlocality*. PhD thesis. 2016. URL: <https://archive-ouverte.unige.ch/unige:88093>.
- [71] **M. T. Quintino**. *Black box correlations: locality, noncontextuality, and convex polytopes*. MSc thesis. 2012. URL: <https://repositorio.ufmg.br/handle/1843/BUOS-A46HJC>.
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Other publications

- [73] **M. T. Quintino**. “Quantum teleportation beyond its standard form: Multi-Port-Based Teleportation”. *Quantum Views* **5**, 56 (2021).

Teaching details

- **Lectures:** more than 150 hours.
- **Tutorials:** more than 410 hours.
- **M1 research projects:** 46 hours of supervision.
- **Course development:** created or substantially redesigned “Quantum Circuits and Query Complexity” (previously “Quantum Circuits and Logic Gates”) and “Quantum Information Overview” (previously “QIIntro”) for Sorbonne University’s [Master’s programme in Quantum Information](#).
- **Summer school:** invited lecturer at [QALYPSO 2026](#), with 2 lectures on semidefinite programming for quantum information.

Recent teaching at Sorbonne University

Undergraduate computer science

- Mathématiques Discrètes (tutorials; 3 semesters).
- Introduction à la calculabilité et à la décidabilité (lectures and tutorials; 4 semesters).

Master's programme in Quantum Information

- Quantum Circuits and Query Complexity (lectures and tutorials; 3 semesters).
- Quantum Information Overview (QIOv, previously QIIntro; lectures and tutorials; 3 semesters).
- M1 research projects (supervision, 2025–2026).
- Quantum Kinematics for Computer Scientists (lectures and tutorials; 1 semester).
- QIT: Quantum Information Theory (lectures and tutorials; 3 semesters).

Other master's programmes

- Théorie de Shannon classique et quantique (lectures and tutorials; 2 semesters).
- Analyse d'algorithmes et génération aléatoire (tutorials; 1 semester).

Teaching history

In semester dates, /1 denotes the first half of the calendar year and /2 the second half. At Sorbonne University, 2026/1 denotes the spring semester of 2026; 2026/2 denotes the autumn semester, which may extend into January 2027.

L2 and L3 denote undergraduate years 2 and 3; M1 and M2 denote master's years 1 and 2.

Sorbonne University, Paris, France

Master d'informatique, Information Quantique (IQ)

2026/1	Quantum Information Overview (QIOv, previously QIIntro) , M1 Lectures (15 h), Tutorials (6 h); UM4INQ51
2026/1	Quantum Information Group Project (PQIG) , M1 M1 research project supervision (20 h); UM4INQ55
2026/1	Advanced Quantum Information Short Project (PQIAS) , M1 M1 research project supervision (8 h); UM4INQ56
2025/2	Quantum Circuits and Query Complexity , M1 Lectures (20 h), Tutorials (22 h); UM4INQ05
2025/1	QIIntro , M1 Lectures (20 h); MU4INQ51
2025/1	Projet QI Moyen (6 ECTS) , M1 M1 research project supervision (12 h); MU4INQ56
2025/1	Projet QI Long (9 ECTS) , M1 M1 research project supervision (6 h); MU4INQ57
2024/2	QIT: Quantum Information Theory , M2 Lectures (4 h), Tutorials (2 h); MU5PYQ03
2024/2	Quantum Circuits and Logic Gates , M1 Lectures (10 h), Tutorials (8 h); MU4INQ05

Sorbonne University — continued

- 2024/1 **QIIntro**, M1
Lectures (4 h), Tutorials (4 h); MU4INQ51
- 2023/2 **QIT: Quantum Information Theory**, M2
Lectures (4 h), Tutorials (2 h); MU5PYQ03
- 2023/2 **Quantum Circuits and Logic Gates**, M1
Lectures (10 h), Tutorials (4 h); MU4INQ05
- 2022/2 **Quantum Kinematics for Computer Scientists**, M1
Lectures (2 h), Tutorials (16 h); MU4INQ01
- 2022/2 **QIT: Quantum Information Theory**, M2
Lectures (6 h), Tutorials (4 h); MU5PYQ03

Licence d'informatique

- 2026/1 **Introduction à la calculabilité et à la décidabilité**, L3
Lectures (16.5 h), Tutorials (3.5 h); LU3IN030
- 2025/1 **Introduction à la calculabilité et à la décidabilité**, L3
Tutorials (38.5 h); LU3IN030
- 2024/2 **Mathématiques Discrètes**, L2
Tutorials (38.5 h); LU2IN005
- 2024/1 **Introduction à la calculabilité et à la décidabilité**, L3
Lectures (1.75 h), Tutorials (38.5 h); LU3IN030
- 2023/2 **Mathématiques Discrètes**, L2
Tutorials (40.25 h); LU2IN005
- 2023/1 **Introduction à la calculabilité et à la décidabilité**, L3
Tutorials (38.5 h); LU3IN030
- 2022/2 **Mathématiques Discrètes**, L2
Tutorials (36.5 h); LU2IN005

Master Sciences, Technologies, Santé

- 2024/1 **Théorie de Shannon classique et quantique**, M1
Lectures (18 h), Tutorials (6 h); MU4PY223
- 2023/1 **Théorie de Shannon classique et quantique**, M1
Lectures (15 h), Tutorials (9 h); MU4PY223

Master d'informatique, Science et Technologie du Logiciel (STL)

- 2022/2 **Analyse d'algorithmes et génération aléatoire**, M2
Tutorials (14 h); MU5IN550

4EU+ European University Alliance

[Sorbonne University, Paris \(online\), France](#)

- 2026/1 **4EU+: Quantum Information and Quantum Many-Body Theory**
Lectures (2 h)
[University of Copenhagen \(online\), Denmark](#)
- 2024/1 **4EU+: Quantum Information and Quantum Many-Body Theory**
Lectures (2 h)

UNESP, Guaratinguetá (online), Brazil

Semana da Física 2021

- 2021 *Não-localidade de Bell: como o indeterminismo quântico permite correlações supra-clássicas*
1 lecture (1 h)

The University of Tokyo, Tokyo, Japan

Undergraduate course in quantum technology

- 2019 *What do you mean by random? Randomness, Quantum Mechanics, and Bell Non-locality*, Undergraduate
1 lecture (~1.75 h)

University of Geneva, Geneva, Switzerland

- 2016/1 **Quantum Information Theory**, Master's
Teaching assistant; tutorials (~16 h)
- 2015/2 **Méthodes mathématiques pour physiciens I**, Undergraduate
Teaching assistant; tutorials (~16 h)
- 2015/1 **Quantum Information Theory**, Master's
Teaching assistant; tutorials (~16 h)
- 2014/1 **Quantum Information Theory**, Master's
Teaching assistant; tutorials (~16 h)
- 2013/1 **Quantum Information Theory**, Master's
Teaching assistant; tutorials (~16 h)

IMPA, Rio de Janeiro, Brazil

28º Colóquio Brasileiro de Matemática

- Jul 2011 **Mecânica quântica para matemáticos em formação**
Teaching assistant

UFMG, Belo Horizonte, Brazil

XXII Escola de Inverno

- 2011 *Comunicação via qubits*
1 lecture (2 h)

Quantum Mechanics 1 (master's course)

- 2010 *Desigualdades de Bell, uma introdução à não-localidade quântica*
1 lecture (1.5 h)